SLID-CH Project Report

Slip Detection in a Compliant Robotic Hand Using Fingertip IMU Data

1. Project Objective

Detect slip / no-slip conditions of a two-finger compliant robotic hand manipulating objects, using fingertip IMU sensor data (3-axis acceleration + 3-axis gyroscope per finger) and classical machine learning.

2. Pipeline Overview

```
SLID-CH RAW DATASET
        |
        v
1. DATA UNDERSTANDING   -- data_understanding.py
        |
        v
2. DATA CLEANING        -- data_cleaning.py
        |
        v
3. DATA PREPARATION     -- prepare_dataset.py
        |
        v
649,928 ML-ready windows
        |
        v
4. EDA + PREPROCESSING  -- eda_preprocessing.py   * CURRENT STAGE *
        |
        v
5. CLASSICAL ML         -- NEXT STEP (ml_models.py)
```

The raw dataset (IMUSlip/) is treated as read-only and has not been modified at any stage.

3. Stage-by-Stage Summary

3.1 data_understanding.py -- Initial Inspection

Purpose: answer "what exactly is inside the raw dataset?" before any processing.

- Inspected 195 IMU trial files (IMU_data.pkl).
- Each file contains a Left IMU and Right IMU signal, each with 6 channels: ax, ay, az, gx, gy, gz -> 12 raw sensor channels per trial.
- Confirmed data types, sample counts, and value ranges for the first trial.
- Exploratory only -- no ML processing performed here.

3.2 data_cleaning.py -- Structural & Numerical Validity Check

Purpose: verify every raw trial file is loadable and well-formed before feature extraction.

Check	What it verifies
File loading	.pkl opens via pickle.load()
Top-level structure	Object is a dictionary
IMU existence	Both Left IMU Signal and Right IMU Signal present

Check	What it verifies
Channel existence	All 6 channels present per IMU
Empty channels	No channel has zero samples
Missing values	No NaN
Infinite values	No +inf / -inf
Equal channel lengths	All 12 channels have matching sample counts

Result:

```
Total trials  : 195
Valid trials  : 195
Invalid trials: 0
```

No raw trial required deletion or repair -- clean result for the report.

3.3 prepare_dataset.py -- Raw Time-Series to ML Feature Table

Purpose: convert variable-length raw time-series into a fixed-width tabular dataset suitable for classical ML.

Windowing

WINDOW_SIZE = 30 consecutive samples per window, overlapping by 1 sample -- this is what produces the large number of observations.

Slip labeling

- Contact-point velocity magnitude computed as v = sqrt(vx^2 + vy^2 + vz^2) from available contact points (LP, LD1, LD2, RP, RD1, RD2).
- SLIP_THRESHOLD = 0.06 m/s -> sample is a "slip sample" if contact speed exceeds this.
- Per window: slip_fraction = mean(valid_labels); window labeled slip = 1 if slip_fraction >= 0.5, else slip = 0.

Important: max_contact_speed and slip_fraction are kept for analysis/labeling only -- they must not be used as ML input features (target leakage).

Feature extraction

Per IMU channel (12 total), computed: Mean, Std, Min, Max, Range, RMS -> 12 x 6 = 72 IMU features.

Row composition (80 columns total)

Group	Columns
IMU statistical features	72
Time features	time_start, time_end, window_duration (3)
Analysis/label-derived	max_contact_speed, slip_fraction (2)
Metadata	object, trial (2)
Target	slip (1)

Output: data_processed/slid_ch_features.csv, slid_ch_features.pkl

Result:

```
Trials found       : 195
Trials processed   : 195
```

```
Trials failed      : 0

Total windows      : 649,928
Total columns      : 80

Class distribution:
  No Slip : 354,206  (~54.5%)
  Slip    : 295,722  (~45.5%)

Objects: Pringles Can, Soup Can, Pudding Box, Plastic Cup, Rubix Cube
```

3.4 eda_preprocessing.py -- Cleaning, Statistics & EDA (current stage)

The largest script; covers most of the Data Mining / Data Science coursework requirements in one pipeline:

```
Prepared dataset -> Data understanding -> Missing-value analysis ->
Missing-value handling -> Invalid-value checking -> Duplicate analysis ->
Data reduction -> Statistical analysis -> Outlier analysis ->
Normalization -> Standardization -> Discretization -> Visualization
```

Missing-value analysis

Total missing values across the prepared dataset = 0 -> no imputation was needed. (Imputation logic is still implemented in the script so the pipeline can handle missing data on other dataset versions.)

Invalid-value analysis

Checked all numeric columns for NaN / infinite values -> 0 invalid values found.

Duplicate analysis

```
Rows before = 649,928
Duplicates  = 0
Rows after  = 649,928
```

No exact duplicate rows removed. Cleaned dataset saved as data_processed/cleaned_dataset.csv.

Descriptive statistics

count, mean, std, min, 25%, 50%, 75%, max computed for all feature columns -> reports/descriptive_statistics.csv.

Central tendency & dispersion

Mean, Median, Mode, MAD, Q1, Q3, Quartile Deviation, Variance, Std Dev, Skewness, Kurtosis -> reports/central_tendency.csv.

Outlier analysis (IQR method)

IQR = Q3 - Q1; bounds = Q1 - 1.5*IQR and Q3 + 1.5*IQR.

Outliers identified but not automatically removed -- a large IMU reading may represent genuine slip motion rather than an error. Treated as an analysis result only.

Min-Max normalization

x' = (x - x_min) / (x_max - x_min) -> confirmed range [0, 1]. Saved to data_processed/normalized_dataset.csv.

Z-score standardization

z = (x - mean) / std -> confirmed mean is approx 0, std is approx 1. Stats saved to reports/transformation_statistics.csv.

Note: these transforms were computed globally for EDA/coursework demonstration. For actual ML training, the scaler will be fit on training trials only.

Discretization

- Contact speed -> Low / Medium / High
- Slip fraction -> Very Low / Low / High / Very High
- Saved to data_processed/discretized_dataset.csv.

These discretized variables are for analysis/coursework only -- not ML input (label-derived).

Visualizations produced (8 figures)

#	Figure	Content
1	Dataset Dashboard	Slip/No-slip bar chart, class proportion pie, IMU feature histogram, slip vs no-slip box plot
2	Central Tendency	2x2 histograms with mean/median marked
3	Histograms	Distributions of selected IMU features
4	Frequency Polygon	Slip vs No-Slip distribution comparison
5	Box Plots	No-Slip vs Slip spread across multiple IMU features
6	Scatter Plots	Left ax vs left gx variability; contact speed vs IMU variability
7	Correlation Heatmap	Relationships/redundancy among numerical variables
8	Transformation Comparison	Original -> Min-Max -> Z-score, side by side

4. Progress Checklist

#	Item	Status
1	Dataset understanding	Done
2	Raw-data structural checking	Done
3	Missing-value detection	Done
4	Invalid-value checking	Done
5	Duplicate checking	Done
6	Raw time-series processing	Done
7	Slip-label generation	Done
8	Sliding-window creation	Done
9	Statistical feature extraction	Done
10	Descriptive statistics	Done
11	Central tendency analysis	Done
12	Dispersion analysis	Done
13	Skewness & kurtosis	Done
14	IQR outlier analysis	Done
15	Min-Max normalization	Done
16	Z-score standardization	Done
17	Discretization	Done

#	Item	Status
18	EDA visualizations	Done
19	Processed datasets saved	Done
20	Reports saved	Done

5. Current Project Structure

```
SLID-CH/
|-- IMUSlip/                        (ORIGINAL DATASET, read-only)
|
|-- src/
|   |-- data_understanding.py      Understand raw data
|   |-- data_cleaning.py           Check raw data quality
|   |-- prepare_dataset.py         Raw -> ML feature dataset
|   `-- eda_preprocessing.py       Cleaning + statistics + EDA
|
|-- data_processed/
|   |-- slid_ch_features.csv
|   |-- slid_ch_features.pkl
|   |-- cleaned_dataset.csv
|   |-- normalized_dataset.csv
|   `-- discretized_dataset.csv
|
|-- figures/
|   |-- 01_dataset_dashboard.png
|   |-- 02_central_tendency.png
|   |-- 03_histograms.png
|   |-- 04_frequency_polygon.png
|   |-- 05_boxplots.png
|   |-- 06_scatter.png
|   |-- 07_correlation.png
|   `-- 08_transformation_comparison.png
|
|-- reports/
|   |-- missing_values.csv
|   |-- invalid_values.csv
|   |-- duplicate_analysis.csv
|   |-- descriptive_statistics.csv
|   |-- central_tendency.csv
|   |-- outlier_analysis.csv
|   `-- transformation_statistics.csv
|
`-- .gitignore
```

6. Next Stage -- Classical ML

```
649,928 windows
       |
       v
Select ML features -> 72 IMU statistical features
       |               (excluding time_start, time_end, window_duration,
       |                max_contact_speed, slip_fraction, object, trial, slip)
       v
```

```
Group by trial
       |
       v
Trial-level Train / Validation / Test split
       |
       v
Scale using TRAIN split only
       |
       v
Classical ML models: Logistic Regression, k-NN, Decision Tree, Random Forest
       |
       v
Evaluation
```

Correction noted for the next stage: eda_preprocessing.py currently reports 75 "ML candidate" features, but this set still includes time_start, time_end, and window_duration. The actual model should use only the 72 IMU statistical features, excluding all time fields, max_contact_speed, slip_fraction, object, trial, and slip -- this avoids target leakage.

Next script to build: ml_models.py, starting with the trial-level train/validation/test split -- not model training directly.